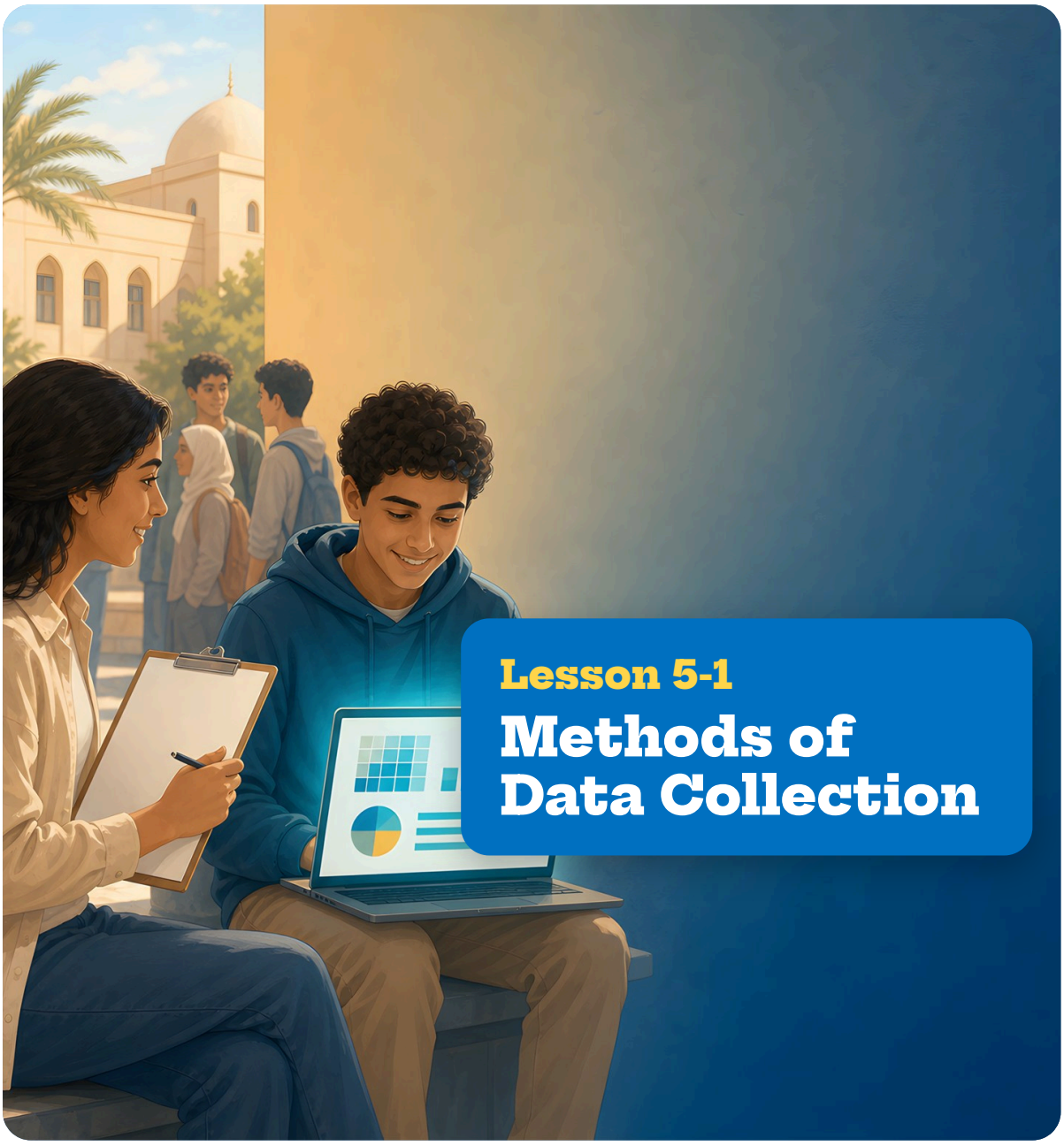


Learning Objectives

- Distinguish primary from secondary data and explain when each is appropriate.
- Explain how sampling works, and describe how sampling bias can lead to confident but wrong conclusions.

Before you can trust what data tells you, you have to trust how it was gathered. Good analysis moves through three stages — collection, organization, then analysis — and if the wrong data is collected, no careful work later can rescue it. Some data you gather yourself for your exact purpose (primary data); some already exists, collected by others (secondary data). And when a whole group is too large to survey, you study a sample — but only a fair sample lets you speak for the whole. A biased sample gives a confident answer that is simply wrong.

? Guiding question: How do you collect data you can trust, and how can a small sample fairly represent a whole population?

Point!

1. The Data Collection Stage

(1) Data analysis progresses through the flow of "collection → organization → analysis," after the purpose has been clarified. In **(data collection)**, the first stage, it is necessary to plan what data will be gathered and how, in line with the purpose of the analysis.

(2) If data that does not match the purpose is collected, it is impossible to draw appropriate conclusions, no matter how carefully later organization and analysis are carried out. For this reason, data collection is an important stage that determines the quality of the entire analysis.

2. Primary Data and Secondary Data

(1) **(Primary data)**…data that is newly collected to fit a specific purpose. *e.g.,* surveys, interviews, observations, experiments. Strength: the items collected can be designed to match the purpose, providing excellent originality and reliability. Constraint: collection takes time, effort, and cost.

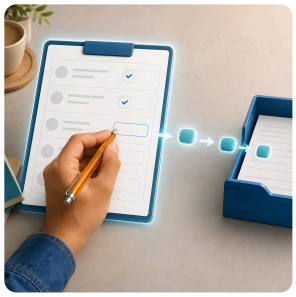
(2) **(Secondary data)**…data that has already been collected and made public by others, and is used in analysis. *e.g.,* open datasets, existing statistics, industry reports. Strength: large amounts of data can be obtained in a short time. Constraints: it may not perfectly match one's purpose; bias or omissions may have arisen during editing or summarizing.

(3) Primary data, since it can be collected to match one's own purpose, has excellent reliability and originality, making it a high-value type of data that can serve as the core of analysis. On the other hand, the value of secondary data lies in being able to easily make use of large-scale data and existing statistics that cannot be obtained from primary data alone. In actual analysis, the two are appropriately combined depending on the purpose and constraints.

Key Fact:
Good analysis starts with good collection: primary data is gathered fresh for a purpose; secondary data already exists; sampling studies part of a population to estimate the whole — and only an unbiased sample supports valid conclusions.

Key Concepts:

- data collection
- primary / secondary data
- population & sample
- sampling (simple random, stratified)
- sampling bias



Sound collection is the base of trustworthy data.

KEY TERMS
data collection — the first stage: plan what to gather and how.
primary data — collected fresh for a purpose.
secondary data — already collected by others.

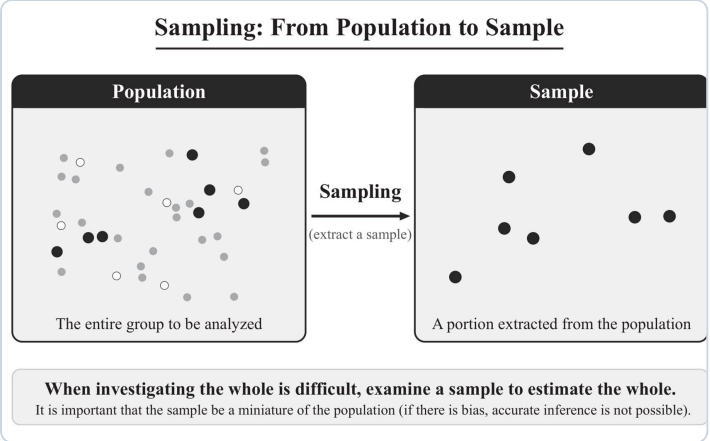


Primary data is gathered fresh; secondary is reused.

3. Sampling

(1) The entire group that is the target of analysis is called the **(population)**, and the part of the population that is actually surveyed is called the **(sample)**. Surveying the entire population is called a **(complete survey)**, also called a census, and surveying a chosen sample is called a **(sample survey)**.

(2) **(Sampling)**…selecting a sample from the population. Depending on the sampling method, the reliability of the conclusions obtained varies greatly.



Sampling: from population to sample

(3) A typical sampling technique is **(random sampling)**, as described below. Random sampling makes it likely that the sample represents the population, allowing the analysis results to be applied to the population as a whole.

Method	Description
(Simple random sampling)	Choose a sample from the population purely by chance, without bias
(Stratified sampling)	Divide the population into strata such as gender or age group, and randomly choose from each stratum

(4) Simple random sampling is the most basic method and can be used even when little is known about the population. Stratified sampling can reflect the composition of the population (such as the ratios of gender and age group) in the sample, making it easier to reduce bias when differences between strata affect the analysis.

4. Bias in Surveys

(1) **(Sampling bias)**…this occurs when the method of selecting a sample is biased and does not correctly reflect the actual state of the population. When sampling bias is present, the conclusions obtained cannot be applied to the population as a whole.

(2) Typical examples of bias. Self-selection and time/location bias concern who is surveyed (selection bias); response bias concerns how those surveyed answer (measurement error).

PAUSE & THINK

A government census you download — is that primary or secondary data for you?



A fair sample mirrors the whole; a biased one misleads.

KEY TERMS

population — the whole group analyzed.
sample — the part actually surveyed.
sampling bias — an unfair sample that misrepresents the whole.

Type of bias	Description	Example
(Self-selection bias)	Bias caused by only those who want to respond actually responding	In a web survey, only people with high interest respond
(Response bias)	Not a selection problem: people already surveyed give answers that look desirable rather than accurate	Reporting a “study time” longer than it really was
(Time/location bias)	Bias caused by the timing or location of the survey being limited	Surveying in front of a station on weekday afternoons, missing office workers and students

(3) It is difficult to eliminate sampling bias completely, but bias can be reduced by carefully designing the survey: how subjects are chosen, how questions are worded, and how the survey is conducted.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A Primary data refers to data that has already been collected and made public by others.

B Selecting a sample from a population is called sampling.

C Stratified sampling is a method in which the population is divided into strata, and a sample is randomly chosen from each stratum.

D When sampling bias is present, the conclusions obtained may not be applicable to the population as a whole.

(2) Match each description (a, b) of a sampling method with the most appropriate option from the choices below (A, B).

a Choose a sample from the population by chance, without bias.

b Divide the population into strata such as gender or age group, and randomly choose from each stratum.

[Choices] **A** Stratified sampling **B** Simple random sampling

✓ Solution

(1)

A Primary data is data that is newly collected to fit a specific purpose. Data already collected and made public by others is secondary data. Therefore, ×.

B Correct definition of sampling. Therefore, ○.

C Correct definition of stratified sampling. Therefore, ○.

D Correct description of the effect of sampling bias. Therefore, ○.

💡 PAUSE & THINK

A web poll anyone can enter — whose voices are missing, and which bias is that?

(2)

a: Choosing without bias by chance is simple random sampling. **B**

b: Dividing into strata and choosing randomly from each is stratified sampling. **A**

Try

1 Answer the following questions.

1. What is the term for data that is newly collected to fit a specific purpose?
2. What is the term for data that has already been collected and made public by others, and is used in analysis?
3. What is the term for selecting a sample from a population?
4. What is the term for when the method of selecting a sample is biased and does not correctly reflect the actual state of the population?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that is most appropriate as a method of collecting primary data.

A Use statistical data published by a government agency.

B Create a survey to fit one's own purpose and collect responses.

C Cite figures from a report issued by an industry association.

D Use survey data from a study conducted earlier by someone else as is.

(2) For each survey situation (a, b), choose the most appropriate sampling method from the choices below (A, B).

a When investigating study time across all students in a school, you want to choose the sample taking into account differences in tendencies between grades.

b You want to grasp the opinions of all eligible voters in a city without bias.

[Choices] **A** Simple random sampling **B** Stratified sampling

EXAM-STYLE QUESTION

Explain why a biased sample can make an analysis worthless even when the calculations are correct. **[6] Refer to:** the sample as a miniature of the population.

PAUSE & THINK

Which option gathers new data *you* designed — and which just reuses others' data?

(3) From the following options (A–D), choose the one that is most appropriate as an example in which sampling bias has occurred.

A Surveying 100 students randomly chosen from all students in the school by drawing lots.

B Dividing all residents into age-group strata and surveying a randomly chosen sample from each stratum.

C Approaching passers-by in front of the station on weekday afternoons and reporting the results as “the opinions of all citizens.”

D Designing the wording of a survey question carefully so that respondents can answer truthfully.

Think as an Engineer — Investigate & Decide

Your school wants to know how long students typically spend commuting to school each day, but you cannot survey everyone. You design how to choose and question a sample.

☐ **Ask whoever is easiest to reach**

☐ **Plan a fair, representative sample**

1 · Collect data (plan the sample). Decide how many students you would ask and how you would pick them so the sample fairly represents the whole school. As a quick pilot, tally the commute of 8–10 classmates. State the population and the sample clearly.

2 · Analyze the risk of bias. Name one group your method might silently drop — for example, students who were absent that day, or students who never open an online survey link. Explain how missing that group would skew the estimated commute, and say which bias it is closest to (self-selection, response, or time/location).

3 · Decide. Choose simple random or stratified sampling for this survey and justify the choice from the risk you found in step 2 — think about whether commute really differs between identifiable subgroups here. **In pairs,** compare methods and agree on the single change that would most reduce bias.

Give evidence for your answer.

Stuck? Start from who might be left out. Ask yourself: could one part of the school answer very differently from the rest — a whole group, or students who never see your survey? If a subgroup might differ systematically, the real question is how to give every subgroup a fair chance *before* you pick your method. Work out which subgroups matter here, then let that guide your choice.

🌍 In a New Context

A shopping mall surveys only visitors on Saturday afternoon and reports the results as “what all our customers think.” Name the bias, explain who is missed, and describe one change that would make the sample fairer.

❓ Lesson Question — Answered

The lesson asked how to collect data you can trust and how a small sample can represent a whole population. Trustworthy analysis begins with collection: after the purpose is clear, you choose primary data (gathered fresh for that purpose, with high reliability but more cost) or secondary data (already collected by others, fast to obtain but perhaps mismatched or biased), often combining both. When the population — the whole group analyzed — is too large, you study a sample through sampling, aiming for a miniature of the population. Random sampling makes that likely: simple random sampling picks purely by chance, while stratified sampling splits the population into strata and picks from each to mirror its make-up. The danger is sampling bias, where a skewed selection cannot speak for the whole; it is reduced by careful design of who is chosen, how questions are worded, and how the survey is run.

KEY TERMS

simple random sampling — picked purely by chance.
stratified sampling — split into strata, then pick from each.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

In data analysis, it is important to plan what data to collect according to the purpose. The data to be collected can be classified into two types: (a), which is newly collected, and (b), which has already been collected and made public by others. Also, when investigating the entire population is difficult, (c) is performed to select a part of the population. At this time, if bias arises in the way of selection, (d) occurs, and the analysis results may not be applicable to the population as a whole.

1. Fill in the blanks (a)–(d).

⭐ REFLECT & CHALLENGE

Reflect: Think of an online poll you have seen. Whose voices were probably missing from it, and why? **Challenge:** You may survey only 40 of your school's 800 students about a new schedule. Describe exactly how you would choose those 40 to avoid bias, and name the one group you would make sure is represented.

3 Answer the following questions.

(1) From the following options (A–D), choose the one that is most appropriate as an example of secondary data.

A Survey responses collected from students in one's own class.

B Population statistics published by the government.

C Traffic on a school commute route observed and recorded by oneself.

D Records of an interview designed and conducted by oneself.

(2) From the following options (A–D), choose the one that most appropriately describes sampling methods.

A Simple random sampling is a method in which the surveyor preferentially chooses subjects that are easy to gather.

B Stratified sampling is a method in which the population is divided into strata, and a sample is randomly chosen from each stratum.

C Simple random sampling is a method in which the population is divided into strata, and a representative is chosen from each stratum.

D A complete survey is a method in which only part of the population is surveyed.

(3) From the following options (A–D), choose the one that is NOT appropriate as a way to reduce sampling bias.

A Choose subjects without bias from the entire population.

B Limit the timing and location of the survey, and gather responses only from people in specific conditions.

C Choose the sample taking into account the strata of the population (age group, gender, etc.).

D Design the wording of survey questions so that respondents can answer truthfully.

⭐ Key takeaway

Remember:
Collect the right data (primary vs secondary), study a fair sample of the population, use random sampling (simple or stratified) so the sample mirrors the whole, and guard against sampling bias.



Learning Objectives

- Explain how missing values, outliers, and duplicate data are handled when cleaning a dataset.
- Describe data type conversion, normalization and standardization, and the reshaping of data to fit a purpose.

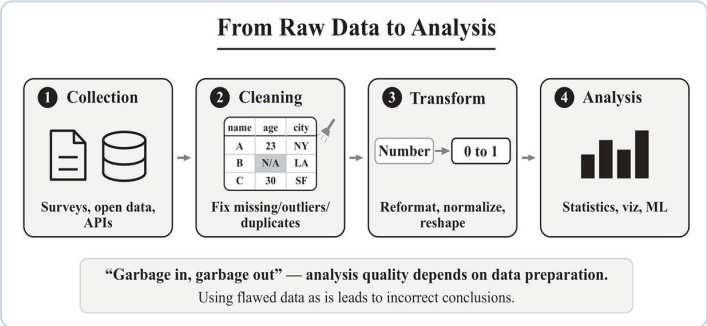
Real data arrives messy: blank cells, a value ten times too large, the same row entered twice, dates written three different ways. Before any analysis can be trusted, the data must be cleaned and transformed to fit the question. There is a rule in computing — "garbage in, garbage out" — which means careful calculation on flawed data still gives a flawed answer. This lesson is about turning raw data into data you can actually rely on.

? Guiding question: Why must data be cleaned and transformed before analysis, and what choices does each fix involve?

Point!

1. The Data Cleaning and Transformation Stages

(1) Collected data often cannot be used for analysis as is. Adjusting issues such as missing values, outliers, and duplicate data, and arranging the data into a format that fits the purpose of the analysis, is called **(data cleaning)** and **(data transformation)**.



From raw data to analysis: collect, clean, transform, analyze

(2) If issues are left unaddressed during analysis, incorrect conclusions may be drawn. For this reason, cleaning and transformation are important stages that support the accuracy and reliability of the analysis.

2. Handling Missing Values

(1) Using data containing missing values (values where data is absent) directly in analysis can prevent calculations from being performed correctly or cause biased results. For missing values, one of the following three responses is chosen.

Key Fact:
Raw data is rarely ready to use. Cleaning fixes missing values, outliers, and duplicates; transformation converts types, rescales, and reshapes data to fit the purpose. "Garbage in, garbage out" — the quality of any analysis depends on this preparation.

Key Concepts:

- data cleaning / transformation
- missing values, outliers, duplicates
- data type conversion
- normalization / standardization
- reshaping (pivot, join, aggregation)

KEY TERMS
data cleaning — fixing missing values, outliers, duplicates.
data transformation — reshaping/rescaling data to fit the purpose.

PAUSE & THINK

A blank cell in a survey — delete the row, fill it in, or flag it? What decides?



Fix blanks and judge outliers before analysis.

Response	Description	When to apply
Deletion	Remove rows or columns containing missing values from the data	When there are few missing values and the impact on the whole is small
Imputation	Fill in the missing parts with values such as the mean, median, or surrounding values	When missing values are scattered without bias, and replacing them with representative values from other data does not significantly change the overall trend
Flagging	Leave a marker indicating that the value was missing, and analyze the data as is	When the missingness itself has meaning (e.g., the fact that someone refused to answer)

(2) Which response to choose is determined by the proportion and pattern of missing values, and the purpose of the analysis. For example, imputing the mean for an item where missing values account for a large proportion of all data may produce values that are far from reality. There is no single correct response; judgment according to the purpose is required.

3. Handling Outliers and Duplicate Data

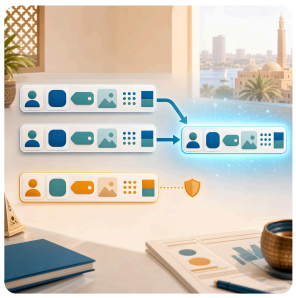
(1) Outliers (values that deviate significantly from other data) are not necessarily errors. They may be **(abnormal values)** caused by input or measurement errors, or they may be values that reflect reality. *e.g.,* If the sales data of a certain store shows that one day has remarkably high sales, it may be an input error, or it may genuinely reflect a sale or event on that day.

KEY TERMS
missing value — a value that is absent.
outlier — a value far from the others.
abnormal value — an outlier caused by an error.

(2) Handling outliers begins with identifying the cause.

① When the cause is an input or measurement error…correct or delete the value

② When it is a value that reflects reality…in principle, keep it and decide how to handle it based on the purpose of the analysis



Remove only genuinely duplicated records.

(3) Duplicate data (a state in which the same content exists in multiple records) affects the results of aggregation and is, in principle, removed. However, it is important to confirm that the data is genuinely duplicated. Even if records appear identical, they may include data that should be retained as separate records. *e.g.,* If the same customer purchases the same product twice on the same day, the records are separate transactions; combining them into one would make the sales aggregation incorrect.

4. Data Type Conversion

(1) Collected data is sometimes difficult to handle in its original format. Arranging the format to make data usable for analysis is called **(data type conversion)**. Typical examples include the following.

Type of conversion	Description	Example
Conversion from string to numeric	Convert numbers stored as strings into numeric values that can be used in calculations	“123” → 123
Unification of date format	Align dates with different notations into a single format	“2026/4/1,” “2026-04-01,” “01 Apr 2026” → 2026-04-01
Unification of notation	Combine values with the same meaning into a single notation	A mix of “Male/Female” and “M/F” → standardized to “Male/Female”

(2) If formats are not aligned, data cannot be aggregated correctly. For example, if numbers are stored as strings, sums and means cannot be calculated; if date notations are not unified, data from the same day will be treated as belonging to different days. For this reason, data type conversion is a step that must be checked before analysis.

5. Normalization and Standardization

(1) When the units or ranges (scales) of data differ greatly between variables, variables with larger values may strongly influence the analysis results. To prevent this, processing is performed to align the scale of data. Typical methods are **(normalization)** and **(standardization)**.

Method	Description	Main purpose
Normalization	Convert data to a range from 0 to 1	Align data with different units into a comparable form
Standardization	Convert data to a form with mean 0 and standard deviation 1	Position values relative to the spread of the data

(2) For example, when analyzing variables with different units and scales together, such as height (cm) and weight (kg), normalization can align both into the range from 0 to 1, making them easier to compare. Specifically, if height data ranges from 160 cm to 180 cm, normalizing converts 160 cm → 0, 170 cm → 0.5, 180 cm → 1, so the minimum becomes 0 and the maximum becomes 1.

(3) Standardization aligns data based on how far each value is from the mean. For example, when test scores from multiple subjects are standardized, even if the mean and spread differ between subjects, the position of each student can be compared.

6. Reshaping Data According to Purpose

(1) Processing that changes the arrangement or grouping of data according to the purpose of the analysis is called **(data reshaping)**. The following three operations are typical.

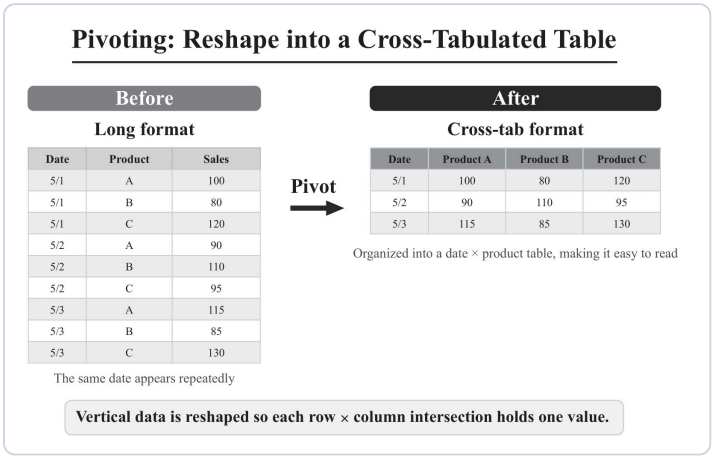
PAUSE & THINK

Height in cm and weight in kg — why might the bigger numbers dominate an analysis?



Normalization rescales variables to a common range.

Operation	Description	Example
(Pivot)	Reshape vertically arranged data into a table of row × column combinations	Reshape data with date, product, and sales arranged vertically into a “date × product” table
(Join)	Connect multiple datasets via a common item	Connect a student roster and a grade sheet by “student ID” into a single table
(Aggregation)	Calculate sums, means, etc. for each group and summarize	Compute monthly totals from daily sales



Pivoting: reshaping long-format data into a cross-tabulated table

(2) Which operation to perform depends on the purpose of the analysis. For example, if you want to see “monthly sales trends,” you need to aggregate daily sales data by month. It is important to clarify the purpose of the analysis first, and then plan how to reshape the data.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A Data cleaning is work performed before data collection.

B The responses to missing values are deletion, imputation, and flagging—three in total.

C All outliers are caused by input or measurement errors, so they must always be deleted.

D Standardization is processing that converts data to a form with mean 0 and standard deviation 1.

(2) Match each description (a–c) with the most appropriate processing from the choices below (A–C).

a Conversion of data to a range from 0 to 1

b Processing that connects different datasets via a common item

c Processing that calculates sums, means, etc. for each group and summarizes

💡 PAUSE & THINK

Is a store's one huge sales day always an error — or could it be real?

[Choices] **A** Aggregation **B** Join **C** Normalization

✓ Solution

(1)

A Data cleaning is the work of arranging collected data, and is performed after collection. Therefore, ×.

B Correct as the responses to missing values. Therefore, ○.

C Outliers include both abnormal values caused by input errors and values that reflect reality. They are not always deleted. Therefore, ×.

D Correct definition of standardization. Therefore, ○.

(2)

a: Conversion to 0–1 is normalization. **C**

b: Connecting via a common item is join. **B**

c: Summarizing by group is aggregation. **A**

Try

1 Answer the following questions.

1. What is the term for the processing that fills in the missing parts of data with values such as the mean or median?
2. What is the term for the processing that converts data to a range from 0 to 1?
3. What is the term for the processing that converts data to a form with mean 0 and standard deviation 1?
4. What is the term for the operation that connects multiple datasets via a common item?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that is NOT an appropriate response to missing values.

A When there are few missing values and the impact on the whole is small, delete rows containing missing values.

B When missing values are scattered randomly, fill in the missing parts with values such as the mean.

C When the missingness itself has meaning, leave a marker indicating that the value was missing and analyze the data.

D Impute the mean and analyze data on items where most of the data is missing.

EXAM-STYLE QUESTION

Explain, using "garbage in, garbage out," why data preparation matters as much as the analysis itself. **[6] Refer to:** one cleaning fix and one transformation.

(2) For each of situations (a, b), choose the most appropriate data processing from the choices below (A, B).

a You want to align date data with mixed notations such as “2026/4/1,” “2026-04-01,” and “01 Apr 2026.”

b You want to compare variables with different units and scales together, such as height (cm) and weight (kg).

[Choices] **A** Data type conversion (standardization of date format)
B Normalization

(3) From the following options (A–D), choose the one that most appropriately describes the handling of outliers.

A Outliers are all errors in the data, so delete all of them as soon as they are found.

B For outliers, identify the cause first; correct or delete them if they are due to input errors, and in principle keep them if they reflect reality.

C Outliers do not affect analysis, so no special handling is needed.

D Outliers can always be resolved by replacing them with the mean.

Think as an Engineer — Investigate & Decide

Your class collected a survey on daily screen time, but the raw table is messy: some cells are blank, one entry reads “9999” hours, dates are written three different ways, and two rows look identical. You prepare the data for analysis.

☐ **Analyze the raw table as is** ☐ **Clean and transform it first**

1 · Inspect the data. List each problem you can see find; for each, say which category it is (missing value, outlier, duplicate, or format issue) and note in one phrase how it would distort the analysis if left unfixed.

2 · Decide a fix and its risk. For the blank cells, choose deletion, imputation, or flagging and justify it from the proportion of blanks. For “9999,” say how you would check its cause — a typo, a wrong unit (minutes entered as hours), or a code entered for a refused answer — before removing it. Name one respondent whose true answer a careless fix might erase.

3 · Plan the transformation. State the one format conversion you must run so the dates aggregate correctly, and whether you would normalize the screen-time column. **In pairs,** compare cleaning plans and agree on which single fix carries the greatest risk of distorting the results — and why.

Give evidence for your answer.

PAUSE & THINK

If most of a column is blank, does filling it with the mean help — or invent data?

KEY TERMS

pivot — long rows → row × column table.
join — link datasets by a common item.
aggregation — summarize by group.

Stuck? Handle one problem type at a time in order: unify formats first, then check the outlier's cause, then decide the blanks, then remove only true duplicates — and never delete a value before you know why it is there.

🌍 In a New Context

A shop's sales table lists each day's sales for products A, B, and C on separate rows, but the manager wants one row per day with a column per product. Name the reshaping operation needed, and name the operation instead required to get each product's monthly total.

❓ Lesson Question — Answered

The lesson asked why data must be cleaned and transformed before analysis and what choices each fix involves. Collected data is rarely usable as is, and flawed data yields flawed conclusions no matter how careful the later calculation — garbage in, garbage out. Cleaning handles missing values (by deletion, imputation, or flagging, chosen from their proportion and meaning), outliers (judged by cause: correct or delete errors, keep real values), and duplicates (removed only when genuinely duplicated). Transformation makes data comparable and analyzable: data type conversion unifies formats (string to numeric, date formats, notation), normalization rescales to 0–1 and standardization to mean 0 and standard deviation 1 so no variable dominates by size, and reshaping — pivot, join, aggregation — arranges data to fit the question. Each step is a judgment guided by the purpose, not a fixed rule.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

If collected data is used directly in analysis, incorrect conclusions may be drawn. For this reason, (a), which addresses issues, and (b), which arranges the format to fit the purpose of the analysis, are necessary. For missing values, an appropriate response is chosen from deletion, (c), and flagging, depending on the proportion of missing values and the purpose of the analysis. Also, when

⭐ REFLECT & CHALLENGE

Reflect: Why can “filling in” missing data sometimes mislead more than leaving it out? **Challenge:** You have one survey column where 70% of answers are blank. Argue whether to impute, flag, or drop the column, and state the one condition under which your choice would change.

units or ranges differ greatly between variables, the scale is aligned through (d), which converts data to a range from 0 to 1, or (e), which converts data to a form with mean 0 and standard deviation 1.

1. Fill in the blanks (a)–(e).

3 Answer the following questions.

(1) From the following options (A–D), choose the one that corresponds to data type conversion.

A Filling in missing values with the mean.

B Standardizing notations such as “2026/4/1” and “01 Apr 2026” into “2026-04-01.”

C Aligning the scales of height and weight to between 0 and 1.

D Aggregating daily sales into monthly totals.

(2) From the following options (A–D), choose the one that most appropriately describes normalization and standardization.

A Normalization is processing that converts data to a form with mean 0 and standard deviation 1.

B Standardization is processing that converts data to a range from 0 to 1.

C Normalization is used to align data with different units into a comparable form.

D Standardization is used to enlarge the scale of variables.

(3) For each processing (a–c), indicate which corresponds to pivot, join, or aggregation.

a Connecting a student roster and a grade sheet by “student ID” into a single table.

b Calculating monthly totals from daily sales data.

c Reshaping daily product sales into the form of a “date × product” table.

[Choices] **A** Pivot **B** Join **C** Aggregation

⭐ Key takeaway

Remember:
Clean first (missing values, outliers, duplicates), then transform (convert types, normalize or standardize, reshape by pivot, join, aggregation). Every fix is a judgment set by the purpose — garbage in, garbage out.



Learning Objectives

- Explain what open data is and identify major reliable sources of it.
- Describe how data is obtained through an API, and explain the importance of terms of use and source citation.

Some of the most powerful data in the world is free to use — population figures from the World Bank, global statistics from the UN, competition datasets on Kaggle. But "free" is not the same as "no rules." To rely on open data you need to know where to get it, how to fetch it (a download in a browser, or an API that returns data directly to your program), and how you are allowed to use it. Checking the terms of use and citing your source are what turn borrowed data into trustworthy analysis.

? Guiding question: Where does reliable open data come from, how do programs fetch it, and what rules govern its use?

📌 Point!

1. Using Open Data

(1) To use open data in analysis, it is necessary to understand where to obtain it, how to obtain it, and how to use it. By correctly observing these points, reliable analysis can be performed.

(2) Open data comes from a variety of sources, such as data from international organizations, data from government agencies, and datasets published by researchers and companies. It is important to choose data sources with high reliability that match the purpose of the analysis.

2. Major Open Data Sources

(1) The following are typical, internationally widely used providers of open data.

Provider	Description	Main uses
(World Bank Open Data)	Statistics on economy, population, education, and health for countries around the world	International comparisons and analysis of social/economic trends in each country
(UN Data)	International statistics published by the United Nations and its related agencies	Analysis of global issues such as population, development, and environment
(Kaggle)	A platform that runs data analysis competitions; datasets in various fields are made public	Acquisition of data for machine learning study and research

(2) On these platforms, data can be searched for and downloaded from a browser, and in many cases data can also be obtained through APIs.

3. Obtaining Data via APIs

(1) **(API (Application Programming Interface))**…rules for how pieces of software interact with each other. In web services, APIs are also widely used as a

Key Fact:
Open data comes from official providers (World Bank Open Data, UN Data) and platforms such as Kaggle, whose reliability varies; get it from a browser or through an API (often returning JSON); every dataset has terms of use to follow and a source to cite.

Key Concepts:

- open data
- World Bank Open Data / UN Data / Kaggle
- API / endpoint / JSON
- terms of use
- source citation

KEY TERMS

open data — data made public for anyone to use.
World Bank Open Data — country statistics on economy, population, and more.



Open data is public and free to reuse.

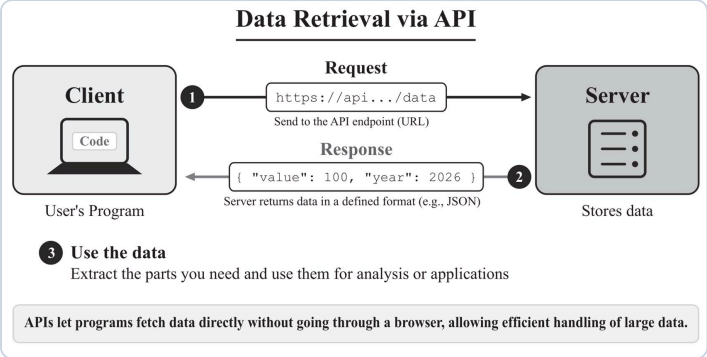
means of obtaining data. By using APIs, necessary data can be obtained directly without going through a browser, allowing large amounts of data to be handled efficiently.

(2) Obtaining data via API proceeds according to the following flow.

① The client (the user's program) sends a request to the API endpoint (the URL that is the destination of the request).

② The server processes the request and returns the requested data in a predetermined format.

③ The client receives the response, extracts the necessary parts, and uses them.



Data retrieval via API: request, response, and use

(3) The data format most widely used for data returned via APIs today is **(JSON (JavaScript Object Notation))**. JSON is a notation that represents data as combinations of names and values; it is easy for humans to read and easy to handle in various programming languages.

(4) When using an API, it is important to **(follow the terms of use)**, not send excessive requests in a short time, and confirm the scope of allowed use of the obtained data.

4. Data Use and Source Citation

(1) Even with open data, the scope of free use is set for each dataset. Always confirm the **(terms of use)** of the data before using it.

(2) When using data, it is required to **(cite the source)**. The source generally includes the data provider, the name of the data, the date of acquisition, and the URL.

(3) Citing the source is important both as an indication of the reliability of the data and to allow others to verify the analysis.

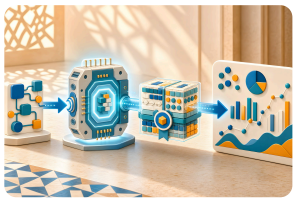
✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A World Bank Open Data publishes statistics on economy, population, education, and so on for countries around the world.

KEY TERMS

API — rules for how software interacts, used to fetch data.
endpoint — the URL a request is sent to.
JSON — a common name/value data format.



An API delivers fresh data straight into analysis.

PAUSE & THINK

"Open" data still has rules — what two things must you check before you use it?



Use data responsibly and credit its source.

B Using an API, necessary data can be obtained directly without going through a browser.

C As long as data is open data, it can be used freely without checking the terms of use.

D When using data, it is required to cite the provider, the name of the data, the date of acquisition, the URL, and so on as the source.

(2) Match each description (a–c) with the most appropriate open data provider from the choices below (A–C).

a Publishes statistics on economy, population, education, and so on for countries around the world.

b Provides international statistics published by the United Nations and its related agencies.

c A platform that runs data analysis competitions.

[Choices] **A** UN Data **B** World Bank Open Data **C** Kaggle

PAUSE & THINK

In an API call, who sends the request and who returns the data?

✓ Solution

(1)

A Correct description of World Bank Open Data. Therefore, ○.

B Correct description of an API. Therefore, ○.

C Even for open data, the terms of use differ for each dataset, so they must always be checked before use. Therefore, ×.

D Correct content of source citation. Therefore, ○.

(2)

a: Statistics on countries around the world is World Bank Open Data. **B**

b: Data from the United Nations and its related agencies is UN Data. **A**

c: The platform that runs data analysis competitions is Kaggle. **C**

EXAM-STYLE QUESTION

Explain why an analyst must check the terms of use and cite the source even when the data is free and open. **[6] Refer to:** your right to use the data, reliability, and letting others verify.

Try

1 Answer the following questions.

1. What is the name of the World Bank's open data portal that publishes statistics on economy, population, education, and so on for countries around the world?
2. What is the term for the rules for how pieces of software interact with each other, which is also widely used in web services for obtaining data?
3. What is the name of the data format widely used when data is returned from an API?
4. What is the term for what should be cited when using data, both to indicate reliability and to allow verification by others?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes the flow of obtaining data via an API.

A The client sends a request to the server, and the server returns the data in a format such as JSON.

B The server one-sidedly continues to send data to the client.

C Clients exchange data directly with each other.

D The server only stores data; clients cannot retrieve it.

(2) For each of purposes (a, b), choose the most appropriate open data provider from the choices below (A–C).

a You want to compare GDP across countries.

b You want to find datasets from various fields for machine learning study.

[Choices] **A** UN Data **B** World Bank Open Data **C** Kaggle

(3) From the following options (A–D), choose the one that most appropriately describes the use of open data.

A Open data can be used freely, so the terms of use do not need to be checked.

B Source citation is required only for data published for a fee.

C Even when using open data, it is required to cite the provider, the name of the data, the date of acquisition, the URL, and so on as the source.

D For data obtained via an API, there is no need to follow the terms of use.

PAUSE & THINK

Which provider fits which purpose — and would you use a browser or an API?

Think as an Engineer — Investigate & Decide

For a school project you want to show how Internet access in Egypt has changed over the past ten years compared with two neighboring countries.

You must choose an open data source and a way to obtain it.

☐ **Take the first dataset you find** ☐ **Choose a reliable source and follow its rules**

1 · Inspect two sources. Look up two candidate providers for this exact question (for example World Bank Open Data and UN Data). For each, note whether it actually holds Internet-access figures by country and year, and how current the latest year is.

2 · Analyze fit and rules. Compare the two on a point the lesson table does not list — such as how far back the data goes, or whether its terms of use allow you to republish a chart in a public report. Name one risk of picking the wrong source here.

3 · Decide. Choose one provider and one fetch method (browser download or API) and justify both from what you found. Write the source citation you would include (provider, data name, date obtained, URL). **In pairs,** compare choices and agree on the more defensible source.

Give evidence for your answer.

Stuck? Start from the question, not the source: list exactly what fields you need (country, year, Internet-access rate), then keep only the providers that actually carry all three — and check each one's terms before you commit.

KEY TERMS

terms of use — the rules set for each dataset.
source citation — provider, name, date, URL.

🌍 In a New Context

A weather app on a phone shows today's forecast by fetching it from a public weather service. Describe the request-and-response steps between the app and the server, and name the data format the server most likely returns.

❓ Lesson Question — Answered

The lesson asked where reliable open data comes from, how programs fetch it, and what rules govern its use. Reliable open data comes from trusted providers chosen to fit the purpose — World Bank Open Data for country economic and social statistics, UN Data for United Nations international statistics, and Kaggle for datasets used in machine learning study. It can be downloaded through a browser or fetched by a program through an API: the client sends a request to an endpoint, and the server returns the data in a set format, most often JSON, which the client then uses. Because open data still comes with rules, every dataset carries terms of use that must be followed — including not overloading an API with requests — and every use requires citing the source (provider, data name, date of acquisition, URL), which signals reliability and lets others verify the analysis.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

To use open data in analysis, it is necessary to understand where to obtain it, how to obtain it, and how to use it. Statistics on economy, population, education, and so on for countries around the world can be obtained from sources such as (a), operated by the World Bank. Also, in addition to downloading individually from a browser, by using (b), necessary data can be obtained directly in formats such as (c) in response to a request. When using open data, the (d) determined for each dataset must always be checked, and when data is used, the provider, the name of the data, the date of acquisition, the URL, and so on must be cited as the (e).

1. Fill in the blanks (a)–(e).

⭐ REFLECT & CHALLENGE

Reflect: Think of a chart you saw online with no source. Did you trust it — and what one line would have changed that?
Challenge: You may cite only one source line for a chart built from a World Bank dataset you downloaded today. Write that citation line, and say which single element you would never drop and why.

3 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes Kaggle.

A A portal operated by the World Bank that publishes statistics for countries around the world.

B A site for international statistics operated by the United Nations.

C A platform that runs data analysis competitions, on which datasets in various fields are made public.

D A system that automatically records browsing histories of websites.

(2) From the following options (A–D), choose the one that is NOT an appropriate description of source citation.

A Source citation is important as an indication of the reliability of the data.

B Source citation is important to allow others to verify the analysis.

C Sources include the data provider, the name of the data, the date of acquisition, the URL, and so on.

D When using open data, it is not necessary to cite the source.

(3) From the following options (A–D), choose the one that is NOT an appropriate point of caution when using an API.

A Use the API in accordance with the terms of use determined by the provider.

B Avoid sending excessive requests in a short time.

C Confirm the scope of allowed use of the obtained data in advance.

D Because of how APIs work, they may be used without checking the terms of use.

⭐ Key takeaway

Remember:
Pick a reliable open data source for your purpose (World Bank Open Data, UN Data, Kaggle), fetch by browser or API (client requests an endpoint; server returns JSON), and always follow the terms of use and cite the source.